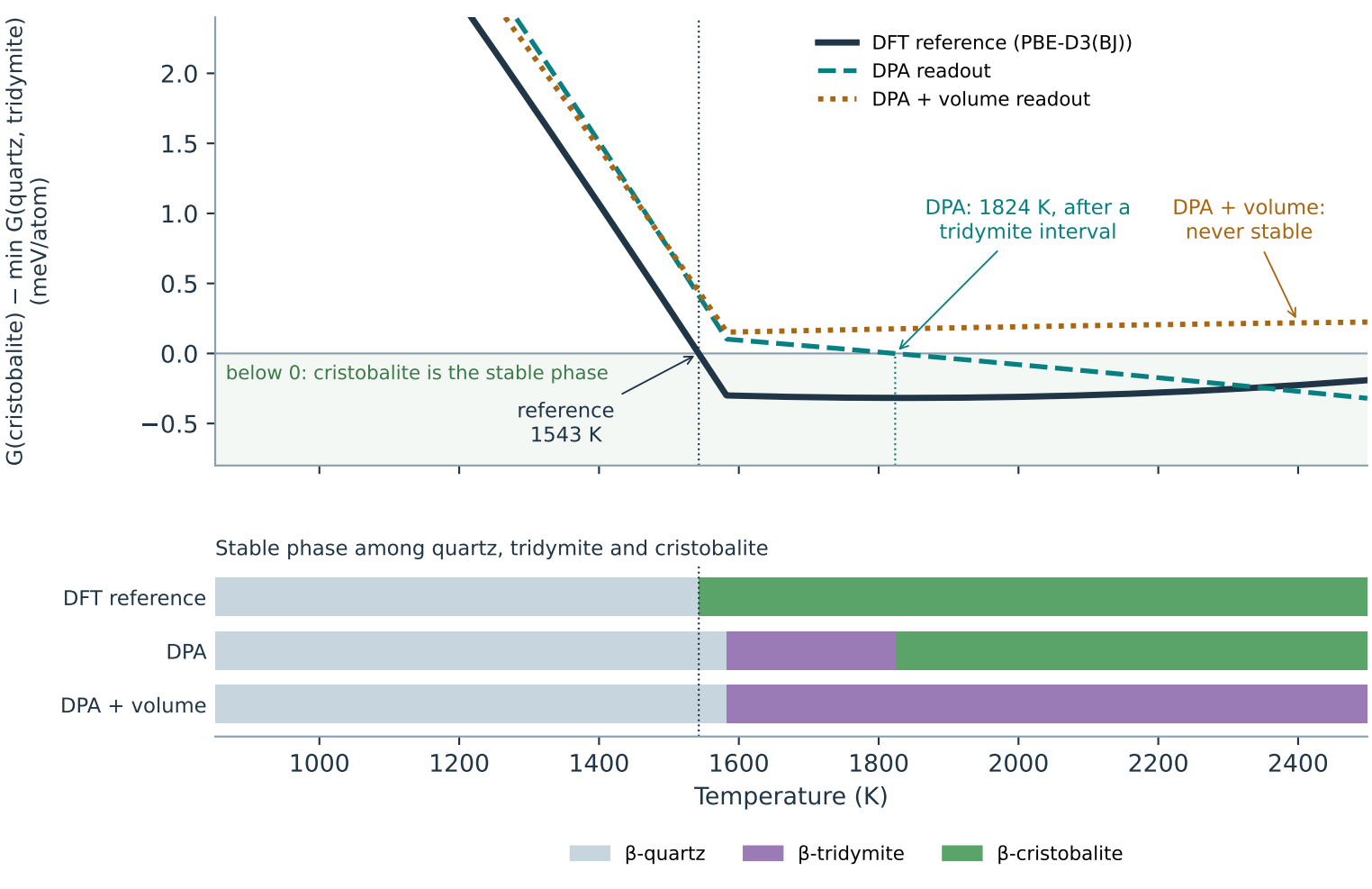
G(cristobalite) − min G(quartz, tridymite) (meV/atom)
2.0
1.5
1.0
0.5
0.0
−0.5
DFT reference (PBE-D3(BJ))
DPA readout
DPA + volume readout
DPA: 1824 K, after a tridymite interval
DPA + volume: never stable
below 0: cristobalite is the stable phase
reference 1543 K
Stable phase among quartz, tridymite and cristobalite
DFT reference
DPA
DPA + volume
1000
1200
1400
1600
1800
2000
2200
2400
Temperature (K)
β-quartz
β-tridymite
β-cristobalite